

Novelty of the RadCluster cluster-dynamics method

Verified against Ghoniem-Org/RadCluster, RadCluster_2.1; benchmarked against Kohnert & Wirth, *Modelling Simul. Mater. Sci. Eng.* **25** (2017) 015008 — September 14, 2026

1 Production bias and simultaneous loop–cavity evolution

The source term. Cascade production (`py_utils/defect_production.py`) is a power-law partition, for $x = i, v$,

$$G_1^x = \eta \dot{d} (1 - f_{\text{cl}}^x), \quad G_n^x = \eta \dot{d} A_x n^{-s_x} \quad (n \geq 2), \quad \sum_n n G_n^x = \eta \dot{d},$$

the monomer channel absorbing the exact remainder so that atom conservation holds even under grid truncation. *Equal total SIA and vacancy inventories are created; they are assigned to different mobility classes.* That asymmetry is the production bias. For the fission spectrum:

Measure	SIA	vacancy	ratio
clustered atom fraction	0.580	0.150	3.87 (i/v)
cluster-number fraction	0.1086	0.0485	2.24 (i/v)
monomer atom fraction	0.420	0.850	2.02 (v/i)
mean produced cluster size	5.34	3.09	—

The “factor of two” is correct for produced *cluster number*, and equivalently for the inverse *monomer* ratio. The clustered atom partition differs by 3.9. Quote whichever is load-bearing, but not “2” for the atom fraction.

What the model does with it. Cascade-born SIA clusters seed loops; the larger vacancy-monomer source feeds cavities. RadCluster evolves both ladders with SIA–SIA and V–V coalescence, residual-size reactions $I_n + V_m \rightarrow I_{n-m}, V_{m-n}$ or $\emptyset$, cavity capture and emission, helium processes, and sink accounting. The size-resolved *coupled* evolution of both families is the substantive advance over a two-monomer swelling model.

The mixed-dimensional cavity kernel (`py_utils/reaction_rates.py`) is

$$\mathcal{K}_{n,m}^{\text{cav}} \propto \frac{m^{1/3} \omega_n^{\text{1D}}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, \quad B_{\text{rot}} = \frac{4}{\pi} \left(\frac{8\pi}{3} \right)^{1/3} \approx 2.627.$$

Do not quote the suppression factor without m . The often-cited $1 + B_{\text{rot}} \hat{L}^2$ is the $m = 1$ reference; larger cavities are progressively *less* suppressed through $m^{-1/3}$. The defensible statement is that *dimensionality changes the relative rates of cavity interception, coalescence and fixed-sink capture* — not that 1D clusters cannot find cavities. Likewise, coalescence reduces mobile cluster number and transfers inventory into large or sessile loops, but it *conserves* SIA mass; it is a redistribution between mobility classes, not a sink.

The exact balance. With balanced production and initial inventories, the code enforces (`py_utils/post_process.py`) $\delta_{\text{FP}} < 10^{-6}$)

$$S_V(t) - S_I(t) = J_I^{\text{sink}}(t) - J_V^{\text{sink}}(t), \quad (1)$$

up to numerical residuals and boundary losses. Internal coalescence and mutual annihilation cancel identically: both destroy or relocate equal numbers of the two species.

Correction: equal capture efficiencies do not zero the right-hand side. An earlier draft of these notes proposed setting $Z_i = Z_v$ to force $\Delta J \rightarrow 0$ and collapse (1) to $S_V = S_I$. **That is wrong.** Equal Z removes *preferential* elastic capture, but the sink fluxes remain unequal because the mobile concentrations and diffusivities of the two species differ — and they differ *precisely because of* the production partition. Equal Z therefore does not force $S_V = S_I$; it removes one cause of the imbalance, not the imbalance.

The decisive design. Isolation requires a 2×2 matrix, not a single run:

	$f_{\text{cl}}^i = f_{\text{cl}}^v$	$f_{\text{cl}}^i > f_{\text{cl}}^v$
$Z_i = Z_v$	reference	production bias only
$Z_i \neq Z_v$	sink bias only	combined (shipped EUROFER deck)

Equalise the dislocation, grain-boundary and precipitate capture efficiencies together (Z^d , Z^{gb} , Z^p — the shipped deck has $Z_i/Z_v = 1.35$, $Z_{i,v}^{gb} = 1.40/0.93$, $Z_{i,v}^p = 1.45/0.93$ at $\rho_d = 5 \times 10^{14} \text{ m}^{-2}$) while holding sink densities, transport, binding, helium conditions and dose rate fixed. The claim is supported if, in the production-only cell, *both* loop and cavity number densities and mean sizes grow relative to the unbiased-source reference. Report the source, coalescence, annihilation, cavity-capture and external-sink fluxes against (1) for each cell.

What is, and is not, new. Golubov–Singh–Trinka rate theory already *permits* production-biased swelling; that is not the novelty. The novelty is the simultaneous *size-resolved* CD evolution of both cluster families under mixed-dimensional transport and nonlocal (residual-size, coalescence) reactions, with the two bias mechanisms separated by controlled construction rather than by asymptotic argument.

2 Bin-moment reduction versus existing grouping techniques

RadCluster stores $\mu_k^{(p)} = \sum_{n \in \mathcal{B}_k} n^p c_n$ and implements

$$\{\mu_k^{(p)}\} \xrightarrow{\mathbf{R}} c_n^{\text{rec}} \xrightarrow{\mathbf{E}} \dot{c}_n^{\text{full}} \xrightarrow{\mathbf{P}} \dot{\mu}_k^{(p)} = \sum_{n \in \mathcal{B}_k} n^p \dot{c}_n^{\text{full}},$$

reconstructing, evaluating the *per-size* master equation, and projecting — at every RHS call (`core/reductions/bin.m`

Kiritani (1973). One average concentration per group with group-averaged rate constants. Inconsistent: densities and their increments are assumed equal within a group, breaking dN/dt and dS/dt .

Golubov *et al.* (2001). Linear intra-group profile $f(x) = L_i^0 + L_i^1(x - \langle x \rangle_i)$ with both moments evolved; number and content are conserved by construction. Not intrinsically nonconservative — the limitations are that it was formulated for *monomer* flux only, and that the intra-group mean flux $\langle J \rangle_i^*$ is undefined at $\Delta x_i = 1$.

Kohnert & Wirth (2017) — the real benchmark. They remove Golubov’s monomer restriction, generalising the moment scheme to arbitrary cluster–cluster reactions, and add four interpolation schemes (linear, secant, Lagrange, cubic spline). Both families work by *projecting the rate constants onto basis functions* to build grouped coefficients $a^\pm, b^\pm$ (their Eqs. 7, 19, 20 and Appendix A), which are “independent of the actual concentration values so they need be computed only once, before time integration begins.”

That precomputation is what buys their speedup, and it carries an explicit price, stated twice in their paper: “*Enforcing constant cluster properties across each group* allows the rate constants to be extracted from the summations” (p. 4), and “due to the assumption that rate constants are unchanging across a group, clusters whose properties differ radically from neighbors in phase space should not be grouped over... we enforce a minimum size to the grouped region... typically around clusters with 20 members” (p. 9).

The distinction. R–E–P forms *no grouped operator at all*, so the constant-property assumption is never made: step **E** evaluates $\mathcal{K}(n, n')$ at the true size. This matters for the physics of §1 specifically, because the fastest-varying kernel in the model is the one carrying the result: $\mathcal{K}_{n,m}^{\text{cav}}$ depends on n through $\omega_n^{\text{ID}} \propto n^{-s}$ and on m through $m^{1/3}/(1 + B_{\text{rot}} \hat{L}^2 m^{-1/3})$, both strongly non-linear across a logarithmic bin. A scheme that holds cluster properties constant over the group flattens exactly the quantity that sets the relative rates of cavity interception and sink capture. **Reaction fidelity, not work reduction, is the advantage.**

A second practical consequence: because no reaction-specific grouped operator is derived, a new channel costs no new coefficient algebra. Kohnert & Wirth report that setup “becomes prohibitive, particularly when applying the moment scheme for problems with dense reaction networks,” requiring computer-algebra-derived analytic forms per reaction class. RadCluster’s loop→network loss and the planned RIS/precipitation edges reuse the existing reduction and conservation machinery unchanged.

	Kiritani (1973)	Golubov (2001)	Kohnert–Wirth (2017)	RadCluster (R–E–P)
State	uniform group density	L^0, L^1 ; linear profile	interpolation nodes <i>or</i> L^0, L^1	constant / linear / log-normal, $P = 1, 2, 3$
Operator	grouped equation	derived group equations	precomputed grouped network	reconstruct grid, per-size RHS, project
Kernels	group approximations	grouped coefficients	grouped; constant across group	original size-dependent kernels
Reactions	monomer	monomer only	arbitrary cluster–cluster	arbitrary; same machinery
New channel	—	re-derive	new analytic coefficients	none required
Conservation	may lose consistency	N, S by construction	moments by construction; interp. leaks $O(\Delta x^2)$	projected stoichiometry, audited
Grouped range	—	—	$\gtrsim 20$ members	no lower restriction
Work	group-level	group-level	group-level (cost ↓ with width)	reduced state, full-grid sweep $I+V$

Qualifications that must accompany the claim. Conservation is *not* unconditional. Linear reconstruction clips negative values (`np.maximum` in `distribution_from_moments_hat`); the discrete log-normal closure normalises $\mu^{(0)}$ exactly but recovers higher moments only approximately; the upper grid boundary can lose products; solver tolerances and floors enter the residual. The correct claim is that RadCluster is *stoichiometrically conservative at the projected-reaction level and continuously audited* via $\delta_{\text{FP}}, \delta_{\text{He}}$. Kohnert & Wirth report the analogous pathologies for their schemes — SDF oscillation, negative concentrations, phase-space-boundary failure — so this is a shared hazard, not a differentiator. Finally, the $O((r-1)^2)$, $O((r-1)^3)$, $O((r-1)^4)$ truncation orders should be removed unless derived for the implemented *discrete* closures, and R–E–P forgoes the asymptotic work reduction that is Kohnert & Wirth’s headline result.

Contribution statement

RadCluster is a production-spectrum-resolved cluster-dynamics framework that couples asymmetric cascade partitioning, mixed 1D/3D SIA transport, SIA–SIA and vacancy–vacancy coalescence, residual-size SIA–vacancy reactions, cavity capture and emission, helium coupling, and explicit sink accounting, so that loop and cavity populations evolve as size-resolved distributions in a single coupled system rather than as separately nucleated averages. On the physics side, controlled calculations that independently equalise cascade clustering fractions and sink capture efficiencies isolate simultaneous loop and cavity growth caused by production bias rather than by preferential sink capture — a separation that reduced-moment rate theory can argue but not demonstrate. On the numerical side, its reconstruct–evaluate–project bin-moment reduction retains the original size-dependent reaction kernels instead of projecting them onto grouped coefficients, so no constant-property assumption is imposed within a bin and no reaction-specific inter-group operator need be derived for new physics; the reduction conserves projected stoichiometric balances under continuous audit of reconstruction, boundary and solver error, trading the asymptotic work reduction of a strict grouped operator for reaction fidelity in exactly the size-dependent, mixed-dimensional kernels that govern the swelling result.